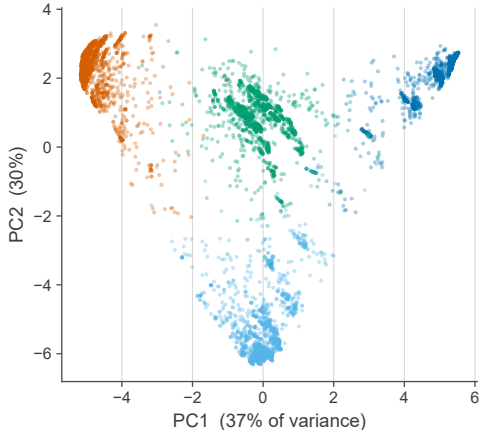


a

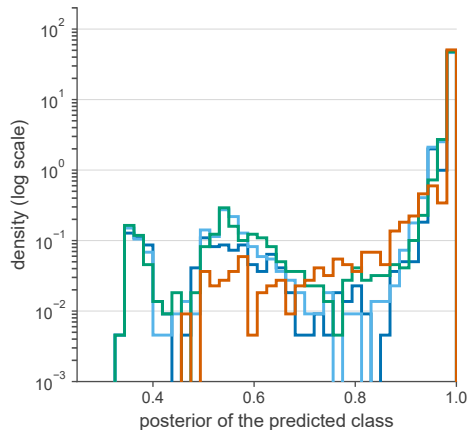
The hidden state separates the rules

● AIIC ● TFT ● WSLS ● AIID

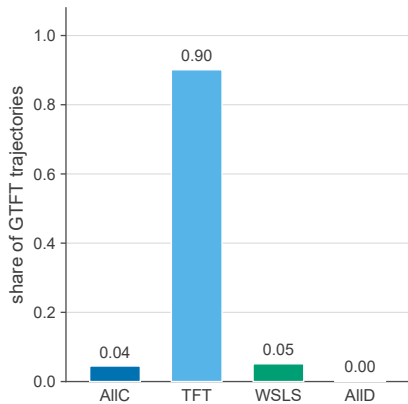
**b**

Almost always certain, on synthetic play

■ AIIC ■ TFT ■ WSLS ■ AIID

**c**

Unseen rule GTFT folds into TFT



Synthetic play only. Panel a is a PCA of the LSTM's final hidden state on 6,000 held-out trajectories, coloured by the strategy that actually generated them -- the network was never asked to make them separable, so the four clusters are a property of the representation. In b the median posterior is 1.00 for all four classes and 95%-97% of trajectories sit above 0.95; that certainty is earned on synthetic play and does not transfer to LLM transcripts, where a fifth of games never reach 0.90 (FR21b). Panel c feeds it generous tit-for-tat, a fifth rule absent from training: 90% is filed as TFT, its nearest true neighbour, and none as AIID. An unseen rule therefore degrades towards its neighbour rather than scattering, which is what makes 'nearest canonical rule' a usable label in FR19.